

SIMATS ENGINEERING ASSESSMENT TOOLS

COURSE NAME: Theory of Computation

COURSE CODE: CSA13

COURSE OUTCOME COVERED

CO4: Construct Context-Free Grammars, generate derivations and parse trees to demonstrate the syntactic structure of strings. (BL : 3)

Assessment Tool Weightage: 10%

QUESTIONS: 10*2=20

Total Marks:20

Data Interpretation

Code of Conduct:

/ PERAM VIVEKANANDA REDDY(192372330)(Name / Reg No) *certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Vivek

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Given Data (Grammar)

Consider the CFG:

$S \rightarrow aSb \mid ab$

Q1. Language Identification

Based on the given grammar, which of the following strings belong to the language?

- A) Ab[Answer]
- B) aabb[Answer]
- C) aaabbb[Answer]
- D) abb

Q2. Pattern Recognition

Consider the CFG:

$S \rightarrow aSb \mid ab$

From the grammar, identify the correct pattern:

- A) $a^n b^n$ [Answer]
- B) $(ab)^n$
- C) $a^n b^m$
- D) $(a+b)^*$

Q3. Derivation Analysis

Which of the following is a valid derivation for aabb?

Consider the CFG:

$S \rightarrow aSb \mid ab$

- A) $S \rightarrow aSb \rightarrow aaSbb \rightarrow aabb$ [Answer]
- B) $S \rightarrow ab \rightarrow aabb$
- C) $S \rightarrow aSb \rightarrow ab$
- D) $S \rightarrow aaSbb \rightarrow aabb$

Q4. Missing Step Identification

Fill in the missing step:

Consider the CFG:

$S \rightarrow aSb \mid ab$ [Answer]

$S \rightarrow aSb \rightarrow aaSbb \rightarrow \underline{\hspace{2cm}} \rightarrow aabb$

- A) ab
- B) S
- C) ϵ
- D) aSb

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Q5. Parse Tree Interpretation

How many levels are there in the parse tree for string aaabbb?

Consider the CFG:

$S \rightarrow aSb \mid ab$

A) 2

B) 3

C) 4

D) 5[Answer]

Consider:

$S \rightarrow SS \mid a$

Q6. Ambiguity Check

This grammar is:

A) Unambiguous

B) Ambiguous[Answer]

C) Regular

D) Deterministic

Q7. String Generation

Consider:

$S \rightarrow SS \mid a$

Which string can be generated?

A) a

B) aa

C) aaa

D) All of the above[Answer]

Q8. Interpretation of Ambiguity

Ambiguity means:

A) One string $\rightarrow$ one parse tree

B) One string $\rightarrow$ multiple parse trees[Answer]

C) No derivation

D) Infinite symbols

Given Data (Grammar 1)

$S \rightarrow 0S1 \mid 01$

Q9. Derivation Interpretation

Which of the following derivations correctly generates the string 0011?

A) $S \rightarrow 0S1 \rightarrow 00S11 \rightarrow 0011$ [Answer]

B) $S \rightarrow 01 \rightarrow 0011$

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C) $S \rightarrow 0S1 \rightarrow 01$

D) $S \rightarrow 00S11 \rightarrow 0011$

Given Data (Grammar 2)

$S \rightarrow aS \mid Sb \mid ab$

Q10. Structural Analysis

Which of the following strings can be generated by the grammar?

A) ab

B) aab

C) abb

D) All of the above **[Answer]**

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RUBRICS FOR CALCULATION

[illegible]